LEARNING MODULE

21

Hypothesis Testing and Multiple Testing

A test asks how surprising the data would be if nothing were there. Run enough tests and something always looks surprising.

VOLUME IV · Statistical Inference and Econometrics

BUILDS ON LM5 · LM13 · LM14 · LM15 · LM16 · LM17 · LM18 · LM19 · LM20

LEADS TO LM22 · LM23 · LM24 · LM25 · LM26 · LM33 · LM34

TIME ≈ 3 h + lab

LEARNING OUTCOMES

Mastery	*After this module you should be able to:*
☐	**1** state a null and an alternative hypothesis, and compute a test statistic, its p-value and the decision at a given level, naming the type I and type II errors
☐	**2** apply the one-sample t-test to a mean return and the two-sample test to two periods or groups of days, state their conditions, and relate a test to a confidence interval
☐	**3** calculate the power of a test of a mean and the years of data needed to detect a given annual Sharpe ratio
☐	**4** test a frequency against a binomial model and the independence of a 2×2 table with the χ^2 test, and explain what dependence between days does to both tests
☐	**5** compute the family-wise error rate and the expected number of false rejections in a family of tests, and apply the Bonferroni correction
☐	**6** apply Holm's procedure and the Benjamini–Hochberg procedure, and state which error each controls and under which condition
☐	**7** explain why the largest t-stat of a search grows with the number of strategies tried, and interpret the t > 3 hurdle
☐	**8** run these tests in Python by hand and with SciPy and statsmodels, checking each function's defaults

LEARNING MODULE OVERVIEW

- A test fixes a null hypothesis, a statistic and a level before the data. The p-value (21.2) is the probability, if the null were true, of a statistic at least as extreme: a probability of data, not of the hypothesis.
- TR23's mean daily return, 0.0625%, has a t-stat of 3.96 by (21.1), a two-sided p-value of 7.6×10^{-5}. Before June 2015 the t-stat is 2.05 and from then 3.68, yet the two means differ by only 1.08 standard errors, (21.3).
- Power grows with the annual Sharpe ratio times the square root of the years, (21.4). At an annual Sharpe ratio of 0.5, 80% power at the 5% level takes 31.4 years of daily data.
- The 2×2 χ^2 equals $n\hat{\rho}^2$, (21.5). TR23's signs pass, with $\chi^2 = 2.69$ and a p-value of 0.10, but falls of 3% or more followed such falls 13 times against 2.0 expected, a p-value of 1.2×10^{-7}. Large moves cluster, and the independent-days null laws fail.
- Wednesday's p-value, 0.018, is the smallest of 17 calendar tests. In such a family 0.85 false rejections are expected, and shuffled dates give a result at least as extreme 0.266 of the time. Bonferroni's adjusted p-value is 0.31, (21.6).
- Holm's rule (21.7) controls the family-wise error rate under any dependence, and Benjamini–Hochberg (21.8) the false discovery rate for independent statistics. On ten constructed p-values they reject 3 and 5, Bonferroni 1; on the calendar none rejects anything.
- The largest of m null statistics grows with m, (21.9): its median is 3.20 at $m = 1{,}000$. The best of 10,000 random long–short rules on TR23 reaches 3.97, and a t-stat of 3 corrects for only about 18.5 tests.

1 INTRODUCTION

LM20 attached a standard error to an estimate. A hypothesis test asks a sharper question: if the true value were zero, how surprising would the estimate be? This module tests a mean, a frequency and a table of counts on TAIEX data, then asks what happens when many tests run on the same data.

MOTIVATING CASE **The best day to own the TAIEX**

From 3 January 2003 to 8 October 2026 the TAIEX total-return index, with dividends reinvested and no costs, gained 0.0625% on an average day. Single days varied far more, with a standard deviation of 1.21%. Is the mean really positive? It lies 3.96 standard errors above zero: 2.05 before the daily price limit widened in June 2015 and 3.68 after it (Exhibit 1).

Now sort the same 5,849 days by calendar month and by weekday, 17 groups in all. Wednesdays stand out. They averaged 0.1332%, against 0.0446% on the other days, a gap of 2.36 standard errors with a p-value of 0.018, well below the usual 5%. Should an investor own the TAIEX only on Wednesdays?

Exhibit 1: TAIEX's mean daily return and its tests

Returns	Days	Mean (% a day)	s_{u} (% a day)	SE (% a day)	t-stat	p-value
All days	5,849	0.0625	1.2085	0.0158	3.956	7.6×10^{-5}
Before 1 June 2015	3,077	0.0464	1.2572	0.0227	2.046	0.041
From 1 June 2015	2,772	0.0804	1.1519	0.0219	3.675	0.00024
Wednesdays minus other days	1,183; 4,666	0.0886		0.0375	2.363	0.018

Note: TR23's simple daily returns. Rows 1 to 3 use (21.1) with $\mu_{\mathrm{null}} = 0$ and standard errors $s_{\mathrm{u}}/\sqrt{\mathrm{days}}$ (LM20's (20.2)); row 4 uses the two-sample statistic (21.3) and its denominator. Row 4's other days include the 23 Saturday sessions (Section 6). Each t-stat uses unrounded inputs; the printed means and s_{u} give 3.955, 2.047 and 3.675. All p-values are two-sided and normal, (21.2). In-sample, 4 of 29 tests on TR23. Source: FinMind `TaiwanStockTotalReturnIndex` (TAIEX), retrieved 18 October 2026. Computations: `data/lm21_numbers.py`.

Sections 2 to 5 test a mean and frequencies, Sections 6 to 8 count the tests, and Section 9 collects the Python.

> **NOTE** **Notation for this module**
>
> The sample is TR23: $T = 5{,}849$ simple daily returns R_t, 3 January 2003 to 8 October 2026, over 23.7645 years (8,680 calendar days from the first close). So $A = 246.12$ sessions a year. $\bar{R}$ and s divide by T, and s_{u}, (19.1), by $T - 1$; every t-stat uses s_{u}. The limit regimes hold 3,077 returns before 1 June 2015 and 2,772 from that date.
>
> Symbols: H_0, H_1, α the hypotheses and the level; μ_{null} the mean under H_0 (μ_0 is LM24's prior mean); θ, θ_0, $\hat{\theta}$ a binomial probability, its null value and a frequency; X a count, x its value; τ years of data; SR the annual Sharpe ratio, $r_f = 0$; subscripts 1 and 2 for two groups ($\bar{R}_1$, μ_1, $s_{\mathrm{u},1}$, T_1); $n_{11}, \ldots, n_{22}$ the cells of a 2×2 table of n pairs (row 1: yesterday down; column 1: today down), $n_{1\cdot}$, $n_{\cdot 1}$ its margins, $\hat{\rho}$ the correlation of its indicators; m tests with m_0 true nulls, ranks i and j, p-values p_i ordered as $p_{(i)}$, q the false-discovery level, i_{H}, i_{BH} the rejection counts of Holm and Benjamini–Hochberg; $Z_1, \ldots, Z_m$ independent standard normal statistics; $f_{i,t} = \pm 1$ rule i's position on day t; V a χ^2 variable. Power and critical values get no letter.

NOTE **Research hygiene**

Every test here is **in-sample**, run on the data that suggested it. The module runs 29 tests on TR23, and the lab lists them: count the tests before you read any single p-value.

Comparing the limit regimes is a **stability check**. Null laws, power curves, shuffled dates and simulated searches are **model** results. The manager is **hypothetical** (LM13), and examples without data are **constructed**. LM16's counts are U28's (28 TWSE sector price indices, 2008–2026), as printed; Section 5 and Problem 8 run two tests on them. Nothing is tested on later data (LM34).

2 WHAT A TEST DECIDES

- [ ] state a null and an alternative hypothesis, and compute a test statistic, its p-value and the decision at a given level, naming the type I and type II errors

EXAMPLE 1

Eight good months in twelve

LM13's hypothetical manager beat the index in 8 of 12 months, independent trials with probability θ of a good month. Is the record evidence of skill, $\theta > \frac{1}{2}$? Find the p-value, the 5% rejection region and its size, and the chance of a rejection at LM13's $\theta = 0.55$.

SOLUTION

Under $H_0 : \theta = \frac{1}{2}$ the number X of good months is $\text{Bin}(12, \frac{1}{2})$, by (15.1). Many good months speak for $H_1 : \theta > \frac{1}{2}$, so the p-value is the upper tail $\Pr(X \geq 8) = 794/4{,}096 = 0.194$. The record is not significant at 5%.

The 5% test rejects when $X \geq x$ for the smallest x with $\Pr(X \geq x) \leq 0.05$. Since $\Pr(X \geq 9) = 0.073$ and $\Pr(X \geq 10) = 79/4{,}096 = 0.0193$, the region is $X \geq 10$. Its size, 0.0193, lies below the level.

For a skilled manager, $\text{Bin}(12, 0.55)$ gives $\Pr(X \geq 10) = 0.042$, so the test misses one with probability 0.958. LM13's posterior of 13.5% answers another question, the probability of skill given the record (LM24).

DEFINITION **Hypothesis test**

A **null hypothesis** H_0 fixes a parameter's value, such as $\theta = \frac{1}{2}$. The **alternative** H_1 says where the value lies otherwise: on one side, $\theta > \frac{1}{2}$ (**one-sided**), or on both, $\mu \neq 0$ (**two-sided**). Both are fixed before the data are seen. A **test statistic** is computed from the data; its **null law** is its distribution under H_0. The test rejects H_0 when the statistic falls in a **rejection region**. Its probability under H_0, the test's **size**, is at most the **level** α.

DEFINITION **p-value**

The **p-value** of an observed statistic is the probability under H_0 of a statistic at least as extreme, in the direction of H_1. The test at level α rejects exactly when the p-value is at most α. For a statistic that is approximately $\mathcal{N}(0,1)$ under H_0, the p-value is

$$\begin{aligned} \text{two-sided:} \quad & 2\Phi(-|\text{t-stat}|), \\ \text{one-sided, } H_1 \text{ above:} \quad & \Phi(-\text{t-stat}). \end{aligned} \tag{21.2}$$

Under a continuous null law the p-value is uniform on $(0,1)$, so $\Pr(\text{p-value} \le u) = u$; this is the inverse transform (18.1) run backwards. A true null therefore gives a p-value below 0.05 one time in twenty.

Two kinds of mistake are possible.

Decision	H_0 true	H_0 false
Reject H_0	type I error, probability at most α	correct, probability = power
Do not reject	correct	type II error, probability = 1 − power

A **type I error** rejects a true null, and a **type II error** keeps a false one. The **power** of a test is its probability of rejecting when a given alternative is true: 0.042 in Example 1. The likelihood-ratio test and the Neyman–Pearson lemma are not in this book (Wasserman 2004, §10.6 and §10.10.1).

EXAMPLE 2

Up days

TR23 has 3,205 up days among its 5,849 returns, and no return is exactly zero. Test $H_0 : \theta = \frac{1}{2}$, a day as likely to rise as to fall, against $H_1 : \theta > \frac{1}{2}$. Treat the days as independent trials.

SOLUTION

The frequency is $\hat{\theta} = 3{,}205/5{,}849 = 0.548$. Under H_0 the count is $\text{Bin}(5{,}849, \frac{1}{2})$, so $\hat{\theta}$ has variance $1/(4T)$ by (15.2). By (17.5) the statistic

$$z = \frac{\hat{\theta} - \frac{1}{2}}{\sqrt{1/(4T)}} = 7.3$$

is approximately standard normal. The exact binomial tail is $\Pr(X \ge 3{,}205) = 1.2 \times 10^{-13}$, so up days beat a fair coin. Both null laws assume independent days, which Section 5 checks.

PITFALL **A p-value is not the probability that H_0 is true**

The p-value is a probability of data given H_0, not of H_0 given the data; LM24 computes the latter, as LM13 did for the manager. "Not rejected" is not "true" either: Example 1's test has power 4%, so it keeps H_0 for most skilled managers. And "significant" is not "large": Wednesday's edge is 0.0886% a day.

導讀 **假設檢定在問什麼**

假設檢定（hypothesis testing）問的是：如果什麼都沒有，資料會有多令人意外？虛無假設（null hypothesis）說「沒有效果」，對立假設（alternative hypothesis）說效果在哪一邊，兩者都要在看資料之前定好。p值（p-value）是在虛無假設成立時，出現至少這麼極端的結果的機率（probability），不是虛無假設為真的機率。p值不超過顯著水準（significance level）就拒絕虛無假設。拒絕了真的虛無假設，叫第一型錯誤（type I error）；沒拒絕假的虛無假設，叫第二型錯誤（type II error）。經理人 12 個月勝 8 個月，p值 0.194，不顯著；就算他真有實力，這個檢定（test）也只有 4% 的機會拒絕。

KNOWLEDGE CHECK 1

A rule produced 60 up days in 100 sessions. Test $\theta = \frac{1}{2}$, two-sided, with the normal approximation. Do you reject at 5%? At 1%?

ANSWER

Here $z = (0.6 - 0.5)/\sqrt{0.25/100} = 2.00$, so the two-sided p-value is $2\Phi(-2) = 0.0455$. Reject at 5%, not at 1%.

3 TESTING A MEAN RETURN

- [ ] apply the one-sample t-test to a mean return and the two-sample test to two periods or groups of days, state their conditions, and relate a test to a confidence interval

LM20 derives $\mathrm{SE}(\bar{R}) = \sigma/\sqrt{T}$, (20.2), and the 95% interval (20.4). This section tests the value that the interval estimates.

RESULT **The one-sample t-test**

Let $R_1, \dots, R_T$ be iid with mean μ and variance $\sigma^2 < \infty$, and test $H_0 : \mu = \mu_{\text{null}}$ with

$$\text{t-stat} = \frac{\bar{R} - \mu_{\text{null}}}{s_{\text{u}}/\sqrt{T}}. \tag{21.1}$$

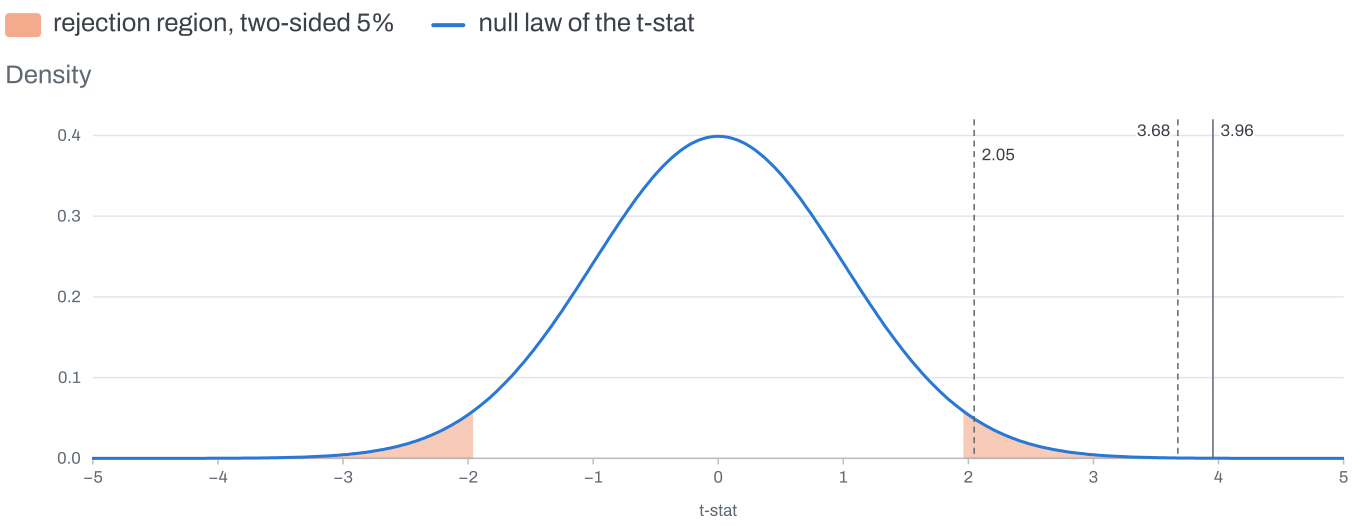
Under H_0, the t-stat is approximately $\mathcal{N}(0,1)$ for large T. If the returns are iid normal, it has exactly the t_{T-1} law for every $T \geq 2$.

The large-T law follows from (17.5), because s_{u} settles at σ (LM20). Here "large" means thousands of days.

The exact law takes one line. For iid normal returns, Blitzstein and Hwang (2019) show that $\bar{R}$ and s_{u}^2 are independent (Example 7.5.9) and that $V := (T-1)s_{\text{u}}^2/\sigma^2 \sim \chi^2_{T-1}$ (Example 10.4.3). With $Z := \sqrt{T}(\bar{R} - \mu)/\sigma$, their Definition 10.4.4 makes $Z/\sqrt{V/(T-1)}$ a t_{T-1} variable. The σ's cancel, so this ratio is (21.1) at $\mu_{\text{null}} = \mu$.

The t_{T-1} law has heavier tails than $\mathcal{N}(0,1)$, but at $T = 5{,}849$ the two nearly coincide (Exhibit 2).

Exhibit 2: The null law of the t-stat and TAIEX's statistics



Note: the standard normal density, the large-sample null law of (21.1); shaded, the two-sided 5% rejection region $|\text{t-stat}| \geq 1.96$. Vertical lines: TR23's t-stats, 3.96 for all days, 2.05 before June 2015 and 3.68 from June 2015 (dashed). The density of t_{5848} differs from φ by at most 2.3×10^{-5} on $[-5, 5]$, and its 97.5% point is 1.9604 against 1.9600, so it is not drawn. The null law is a model; the statistics are in-sample. Source: as Exhibit 1.

EXAMPLE 3

Is TAIEX's mean positive?

Test $H_0 : \mu = 0$ against $H_1 : \mu \neq 0$ on TR23 at the 5% level, with LM19's $\bar{R} = 0.0625\%$ and $s_{\text{u}} = 1.2085\%$ a day.

SOLUTION

The standard error is $s_{\mathrm{u}}/\sqrt{T} = 1.2085\%/\sqrt{5849} = 0.0158\%$, and (21.1) gives t-stat $= 3.96$ from unrounded inputs. By (21.2) the two-sided p-value is 7.6×10^{-5}. The t_{5848} law gives 7.7×10^{-5}, and the one-sided p-value is 3.8×10^{-5}. Reject H_0: the mean is positive at any usual level. With s in place of s_{u} the t-stat is still 3.956 to three decimals, so at this T the divisor does not matter.

LM20's 95% interval for the annual mean, 7.8% to 23.0% a year, excludes 0: the same decision at 5%. With the same standard error, a two-sided test at level α rejects μ_{null} exactly when μ_{null} lies outside $\bar{R} \pm \Phi^{-1}(1-\alpha/2)\,\mathrm{SE}$. An interval is the set of null values that the test keeps.

The conditions matter more than normality. By (16.3), uncorrelated returns give $\operatorname{Var}\bar{R} = \sigma^2/T$ even when volatility clusters, while autocorrelated returns change it. TR23's log returns have a lag-1 autocorrelation of 0.0393 (LM19), and LM23's HAC errors correct for autocorrelation.

RESULT **Two-sample statistic with unpooled variances**

Let two independent groups of T_1 and T_2 returns each be iid, with means μ_1, μ_2 and finite variances. Under $H_0 : \mu_1 = \mu_2$, for large T_1 and T_2,

$$\frac{\bar{R}_1 - \bar{R}_2}{\sqrt{s_{\mathrm{u},1}^2/T_1 + s_{\mathrm{u},2}^2/T_2}} \tag{21.3}$$

is approximately $\mathcal{N}(0,1)$.

The denominator estimates $\sqrt{\operatorname{Var}(\bar{R}_1 - \bar{R}_2)}$, which (16.3) gives for independent groups. SciPy computes (21.3) as `ttest_ind(..., equal_var=False)`, with a t reference law.

EXAMPLE 4

Did the mean change in June 2015?

Before 1 June 2015 TR23's mean was 0.0464% a day, with a t-stat of 2.05; from that date it was 0.0804%, with 3.68. The standard errors are 0.0227% and 0.0219% (Exhibit 1). Treat the regimes as independent groups. Is the difference significant at 5%?

SOLUTION

The standard error of the difference is $\sqrt{0.0227^2 + 0.0219^2} = 0.0315\%$. So (21.3) gives $(0.0804 - 0.0464)/0.0315 = 1.08$, a two-sided p-value of 0.28; SciPy's t reference gives 0.280. One period passes 1% and the other 5%, yet the difference between them is not significant. Stability check.

A change in variance between the regimes needs LM20's standard error of a volatility, and volatility clustering needs LM22's blocks of days. That standard error needs a finite fourth moment, which LM15's Student t fit ($\nu = 3.065$) lacks, so it is nominal. LM20 gives the kurtosis no standard error at all.

PITFALL **Significant in one period, not in the other**

"Significant after 2015, not before" does not show that the mean changed. Test the difference itself: Example 4 gives 1.08.

KNOWLEDGE CHECK 2

A strategy averages 0.04% a day with $s_{\mathrm{u}} = 1\%$ over 900 days. Find its t-stat. With the same mean and s_{u}, how many days give a t-stat of 2?

ANSWER

By (21.1), $0.04/(1/\sqrt{900}) = 1.20$. A t-stat of 2 needs $\sqrt{T} = 2/0.04 = 50$, that is 2,500 days, about ten years.

4 POWER AND THE YEARS A SHARPE RATIO NEEDS

- [ ] calculate the power of a test of a mean and the years of data needed to detect a given annual Sharpe ratio

Let $\mathrm{SR} = \sqrt{A}\,\mu/\sigma$ be the annual Sharpe ratio of iid returns with known σ and $r_f = 0$. Over τ years, $T = A\tau$, so the statistic $\sqrt{T}\bar{R}/\sigma$ is approximately $\mathcal{N}(\sqrt{T}\mu/\sigma, 1)$, and $\sqrt{T}\mu/\sigma = \mathrm{SR}\sqrt{\tau}$. The two-sided test rejects when $|\text{t-stat}| \geq \Phi^{-1}(1-\alpha/2)$, and (5.5) gives the probability of each tail.

RESULT **Power of the test of a mean**

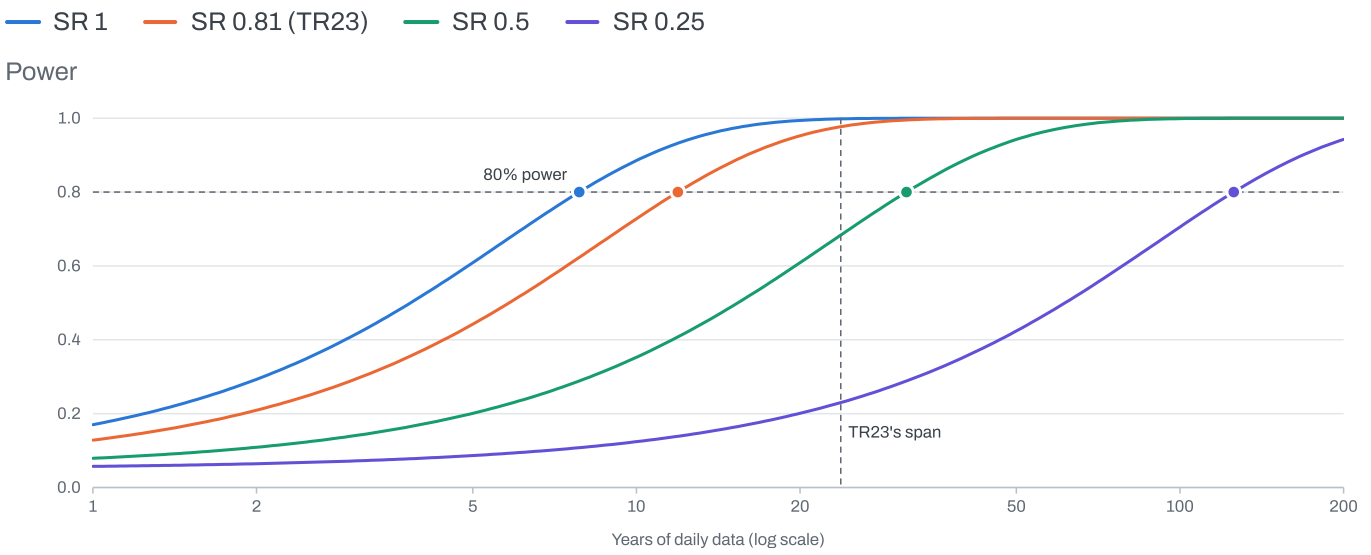
Let returns be iid with known σ and annual Sharpe ratio SR ($r_f = 0$), observed for τ years. At level α, the two-sided test of $\mu = 0$ has

$$\begin{aligned} \text{power} &\approx \Phi\big(\mathrm{SR}\sqrt{\tau} - \Phi^{-1}(1-\tfrac{\alpha}{2})\big) \\ &\quad + \Phi\big(-\mathrm{SR}\sqrt{\tau} - \Phi^{-1}(1-\tfrac{\alpha}{2})\big), \\ \tau &\approx \left(\frac{\Phi^{-1}(1-\frac{\alpha}{2}) + \Phi^{-1}(\text{power})}{\mathrm{SR}}\right)^2, \end{aligned} \tag{21.4}$$

the second when the first line's second term is negligible.

LM19 annualizes a daily Sharpe ratio by $\sqrt{A}$, a convention on simple returns; even for log returns it needs uncorrelated days (Lo 2002). With s in place of σ the identity is exact on data: $\sqrt{T}\bar{R}/s = \mathrm{SR}\sqrt{\tau}$.

Exhibit 3: Power of the two-sided 5% test of a zero mean



Note: (21.4) for annual Sharpe ratios of 1, 0.81 (TR23's), 0.5 and 0.25, against years of daily data on a log axis. Power reaches 80% at 7.85, 11.9, 31.4 and 125.6 years; over TR23's 23.76 years it is 0.998, 0.977, 0.68 and 0.23. Model: iid returns, σ known, $r_f = 0$, annual $\mathrm{SR} = \sqrt{A}\,\bar{R}/s$ with $A = 246.12$. TR23's 0.81 is LM19's, in-sample.

EXAMPLE 5

Years of data for 80% power

How many years of daily data give 80% power at the 5% level against annual Sharpe ratios of 1, 0.5 and 0.25? And against TR23's own, LM19's $\sqrt{246.12} \times 0.0625\%/1.2084\% = 0.81$?

SOLUTION

With $\Phi^{-1}(0.975) = 1.959964$ and $\Phi^{-1}(0.8) = 0.841621$, (21.4) gives $\tau = (2.801585/\mathrm{SR})^2$: 7.85, 31.4 and 125.6 years. TR23's ratio, 0.8115 to four decimals, needs 11.9 years. Its t-stat is $0.8115 \times \sqrt{23.7645} = 3.956$, by the identity above. If 0.81 were the true ratio, TR23's 23.76 years would give power 0.98; at 0.5 the power would be only 0.68 (Exhibit 3). Model.

PITFALL **More frequent data do not buy power**

Power depends on $\mathrm{SR}\sqrt{\tau}$, the span in years, not on the number of observations. Hourly returns over the same 23.76 years would not shrink the standard error of the annual mean; LM20 shows why (Merton 1980).

導讀 **檢定力：為什麼 23.8 年還不夠**

檢定力（power）是對立假設（alternative hypothesis）為真時，檢定（test）拒絕虛無假設（null hypothesis）的機率（probability）。平均報酬（mean return）的 t 統計量（t-statistic）約等於年化夏普比率（annualized Sharpe ratio）乘以資料年數的平方根，所以標準誤（standard error）只隨年數縮小，與每天或每小時抽樣（sampling）無關。在 5% 的顯著水準（significance level）下要有 80% 的檢定力，夏普比率為 0.5 需要 31.4 年的日資料，0.25 更需要 125.6 年。加權股價報酬指數有 23.8 年資料，對 0.5 的檢定力也只有 0.68。回測（backtest）幾年就宣稱找到優勢，多半還分不出真假。

KNOWLEDGE CHECK 3

At an annual Sharpe ratio of 0.5, what power does the two-sided 5% test have with ten years of data? How many years make the expected t-stat equal to 3 at $\mathrm{SR} = 1$ and at $\mathrm{SR} = 0.5$?

ANSWER

By (21.4), the power is

$$\Phi(0.5\sqrt{10} - 1.96) + \Phi(-0.5\sqrt{10} - 1.96) = 0.35.$$

The expected t-stat is $\mathrm{SR}\sqrt{\tau}$, so 3 needs $\tau = (3/\mathrm{SR})^2$: 9 years at $\mathrm{SR} = 1$ and 36 at 0.5.

5 TESTING FREQUENCIES AND INDEPENDENCE

- ☐ test a frequency against a binomial model and the independence of a 2×2 table with the χ^2 test, and explain what dependence between days does to both tests

LM13 counted 1,226 down days after a down day, against 1,195.0 under independence: 31 days in 5,848. Is the gap larger than chance? Exhibit 4 tests it and one more gap.

Exhibit 4: Two tests of independence on TAIEX's day pairs

Day t after day $t-1$	Trials	Observed	Expected	Test	p-value
Down after down	5,848 pairs	1,226	1,194.95	$\chi^2 = 2.69$	0.10
Fall of 3% or more after one	108 days	13	2.0	exact binomial	1.2×10^{-7}

Note: LM13's 5,848 pairs of adjacent days; down means $R < 0$, a fall $R \le -3\%$. Expected counts keep both days' margins by (13.6) and are exact fractions (LM13 printed 1,195.0). Row 1 uses (21.5) with the χ^2_1 law; row 2 the one-sided upper tail of $\mathrm{Bin}(108, 108/5{,}848)$. Both null laws assume independent days, which row 2 rejects. In-sample. Source: as Exhibit 1.

RESULT **The chi-square test of a 2×2 table**

Classify n pairs by two indicators, with cells n_{11}, n_{12}, n_{21}, n_{22}, row totals $n_{1\cdot}$, $n_{2\cdot}$ and column totals $n_{\cdot 1}$, $n_{\cdot 2}$. By (13.6), the expected count of a cell under independence is its row total times its column total over n. Then

$$\chi^2 := \sum_{\text{cells}} \frac{(\text{observed} - \text{expected})^2}{\text{expected}} = n\hat{\rho}^2, \tag{21.5}$$

with $\hat{\rho}$ the correlation of the two indicators over the n pairs. If the pairs are iid and the two indicators independent, χ^2 is approximately χ^2_1 for large n.

For one series on consecutive days, $\hat{\rho}$ is, up to its first and last day, LM19's lag-1 autocorrelation $\hat{\rho}_1$ of the indicator series. Under independent days $\sqrt{T}\hat{\rho}_1$ is approximately standard normal; this is LM19's band (19.6). So $n\hat{\rho}^2$ is approximately a squared standard normal, which is χ^2_1 by LM16's definition. For two series on the same days, the χ^2_1 law is stated without proof.

DEEP DIVE **Why the 2×2 χ^2 is $n\hat{\rho}^2$** optional · click to expand

Let $D := n_{11}n_{22} - n_{12}n_{21}$. Cell (1,1) has observed minus expected $n_{11} - n_{1\cdot}n_{\cdot 1}/n = D/n$, because $n\,n_{11} - (n_{11}+n_{12})(n_{11}+n_{21}) = D$. The fixed margins force the other three cells to $\pm D/n$ as well. The reciprocals of the four expected counts sum to

$$\frac{n}{n_{1\cdot}n_{\cdot 1}} + \frac{n}{n_{1\cdot}n_{\cdot 2}} + \frac{n}{n_{2\cdot}n_{\cdot 1}} + \frac{n}{n_{2\cdot}n_{\cdot 2}} = \frac{n^3}{n_{1\cdot}n_{2\cdot}n_{\cdot 1}n_{\cdot 2}},$$

since $n_{1\cdot} + n_{2\cdot} = n_{\cdot 1} + n_{\cdot 2} = n$. Hence $\chi^2 = nD^2/(n_{1\cdot}n_{2\cdot}n_{\cdot 1}n_{\cdot 2})$. The covariance of the two indicators is D/n^2, and their variances are $n_{1\cdot}n_{2\cdot}/n^2$ and $n_{\cdot 1}n_{\cdot 2}/n^2$ (LM16). So $\hat{\rho} = D/\sqrt{n_{1\cdot}n_{2\cdot}n_{\cdot 1}n_{\cdot 2}}$, and $\chi^2 = n\hat{\rho}^2$.

EXAMPLE 6

Down after down

Test the independence of TR23's signs on LM13's 5,848 pairs. There were 1,226 downs after a down; yesterday's margins are 2,643 down and 3,205 up, and today's 2,644 and 3,204.

SOLUTION

The expected counts are 1,194.95, 1,448.05, 1,449.05 and 1,755.95, as exact fractions. Each observed count differs from its expected count by 31.05. Then (21.5) gives $\chi^2 = 2.69$, or $5{,}848 \times 0.02143^2$, with $\hat{\rho} = 0.0214$, LM15's ρ_1. Its χ^2_1 upper tail is a p-value of 0.10: no evidence against independent signs at 5%. SciPy's `chi2_contingency` applies Yates' continuity correction by default and prints 2.60, with a p-value of 0.107.

EXAMPLE 7

Falls after falls

On the pair space, 108 days follow a fall of 3% or more, and 13 of them fell 3% or more too (LM13). Test whether such falls are independent of the day before.

SOLUTION

Under independence each of the 108 days falls with probability $\theta_0 = 108/5{,}848$. The count is then $\mathrm{Bin}(108, \theta_0)$, with mean $108^2/5{,}848 = 2.0$. The exact one-sided tail is $\Pr(X \geq 13) = 1.2 \times 10^{-7}$. Large moves cluster, so the days are not independent. The null laws of Example 2 and Example 6 hold for the signs only approximately.

Clustering of large moves breaks the count tests of extreme days, such as Example 7's and LM16's. A count is a sum of indicators, and (16.3) adds twice their covariances to their variances. Clustered extreme days make those covariances positive, so the count spreads more than the binomial says and the test rejects too often. LM22's block bootstrap is the tool for this, and LM26 models the clustering.

On U28, LM16 found 177 days outside the normal model's whitened circle of radius 3, against a binomial mean of 48.68 and standard deviation of 6.94. From those printed counts, $z = (177 - 48.68)/6.94 = 18.5$. Clustering would have to widen the spread 6.2-fold, to 42.8 days, to bring z down to 3.

LM16's five-sector counts read as 4,382 tests on χ^2_5 at the 1% level, one a day: 43.8 rejections expected, 291 found. Its crash correlation, 0.788 against the model's 0.860, needs the sampling law of a conditional correlation, which LM22's block bootstrap estimates.

PITFALL **Two-proportion tests on clustered days**

Comparing the share of extreme days in two regimes, as LM16's lab does, invites a test of two proportions. That test treats clustered days as independent, so its p-value is too small; LM22's block bootstrap is the tool for this.

KNOWLEDGE CHECK 4

In 10,000 day pairs, how large must $|\hat{\rho}|$ be for the 2×2 χ^2 to exceed the χ^2_1 5% point, 3.84?

ANSWER

By (21.5), $10{,}000\,\hat{\rho}^2 > 3.84$ needs $|\hat{\rho}| > 0.0196$, which is $1.96/\sqrt{n}$. The χ^2 test and the two-sided band (19.6) agree.

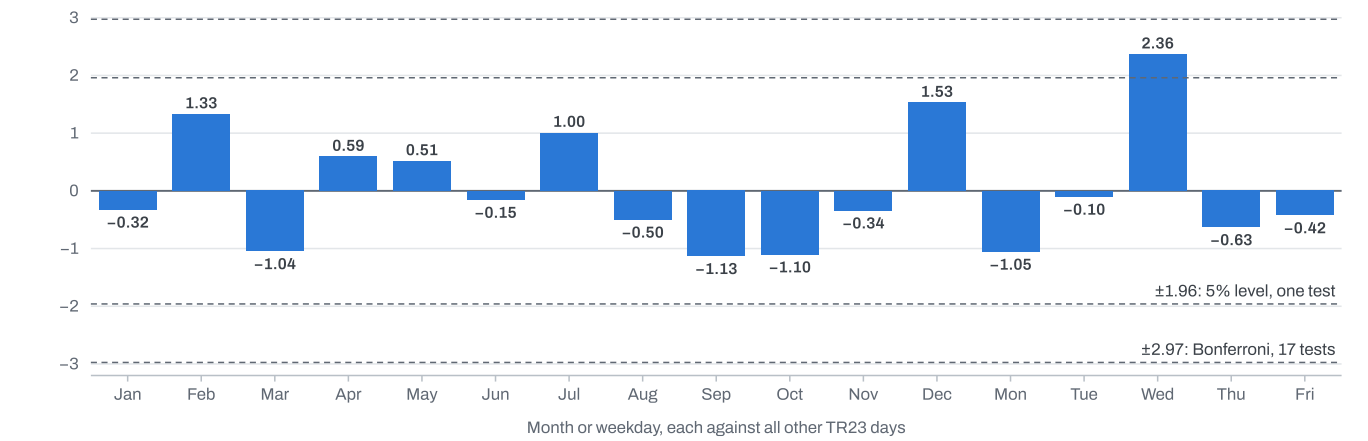
6 MANY TESTS AT ONCE

- ☐ compute the family-wise error rate and the expected number of false rejections in a family of tests, and apply the Bonferroni correction

Wednesday was not a hypothesis fixed in advance. It was found by sorting the days, so its p-value belongs to a family: 12 months and 5 weekdays, each tested against all other TR23 days with (21.3). That makes 17 tests (Exhibit 5).

TR23 also holds 23 Saturday sessions (2006–2018), which belong to no weekday and to every "rest". TWSE's history page states that the market stopped trading on Saturday make-up workdays from 2019; that the 23 sessions were such days is an inference.

Exhibit 5: Seventeen calendar tests on TAIEX



Note: the two-sample statistic (21.3) of each month and weekday against all other TR23 returns. Months hold 450, 369, 532, 477, 503, 498, 524, 522, 491, 482, 492 and 509 days (January to December). Weekdays hold 1,151, 1,181, 1,183, 1,168 and 1,143 (Monday to Friday). Dashed lines: ±1.96, the 5% level of one test, and ±2.97, Bonferroni's level for 17 tests. In-sample. Source: as Exhibit 1.

If every null were true, (14.1) and (14.3) would make the family expect $17 \times 0.05 = 0.85$ false rejections at 5%, whatever the dependence. One test rejects: Wednesday's, at 2.36. February (0.1405% a day, 1.33) and December (0.1270%, 1.53) have the highest monthly means (Exhibit 5).

The **family-wise error rate** (FWER) is the probability of at least one false rejection in the family. For 17 independent tests of true nulls at 5%, (13.7) gives $1 - 0.95^{17} = 0.58$.

RESULT **Bonferroni's correction**

Let m tests include m_0 true nulls, and run each test at level α/m. Then, under any dependence,

$$\text{FWER} \leq m_0 \frac{\alpha}{m} \leq \alpha. \tag{21.6}$$

For m independent tests of true nulls, each of size α (as under a continuous null law), $\text{FWER} = 1 - (1 - \alpha)^m$. Under any dependence, the expected number of false rejections is the sum of the true nulls' sizes, at most m_0 times each test's level.

The bound is (13.2) over the m_0 events "true null i is rejected", each of probability at most α/m. The independent case is (13.7), and the expected count is (14.1) with (14.3). For the calendar, Bonferroni's level is $0.05/17 = 0.00294$, or $|z| \geq 2.974$. Equivalently, multiply each p-value by m and compare with α.

EXAMPLE 8

Wednesday in its family

Wednesday's p-value, 0.018, is the smallest of the 17. How surprising is that in a family of 17? Answer with Bonferroni, for independent tests and by shuffling TR23's dates.

SOLUTION

Bonferroni's adjusted p-value is $17\,p_{(1)} = 0.31$. Šidák's adjustment for independent tests is $1 - (1 - p_{(1)})^{17} = 0.267$. Neither is below 5%.

The tests are dependent, because the months partition the days, and so do the weekdays. Experiment 8 shuffles the returns over TR23's fixed dates, so every group keeps its size. Each shuffle is `R[perm]`, with `perm = rng.permutation(T)` drawn 10,000 times from one `rng = default_rng([8, 20261011])`. Across the shuffles, 0.580 ± 0.005 of the families have a test at $|z| \geq 1.96$, against 0.582 for independent tests. A largest statistic at least as extreme as Wednesday's 2.36 occurs in 0.266 ± 0.004 of them, with standard errors by (18.2). Model.

Stability check: within each regime, Wednesdays against the other days give 2.06 before June 2015 (p-value 0.040) and 1.25 after (0.21).

Here the shuffle is only a simulated null, and permutation-test theory is not in this book; LM22 reads the shuffle as resampling without replacement.

This module runs 29 tests on TR23: seven in Sections 2 to 5, the 17 of the calendar, Wednesday in each regime and three in Problems 7 and 9. Report every p-value with its family and the number of tests run on the same data.

導讀 **一次做很多個檢定**

多重假設檢定（multiple testing）的問題是：檢定（test）做得越多，純屬偶然的「顯著」（significant）結果就越多。17 個檢定各在 5% 下進行，若虛無假設（null hypothesis）全為真，平均會有 0.85 個錯誤拒絕（false rejection）。若檢定彼此獨立（independent），至少錯一次的機率（probability），也就是族系誤差率（family-wise error rate），高達 0.58。星期三的 p值（p-value）0.018 是 17 個裡最小的，放回這 17 個檢定之中就不稀奇：打亂日期之後，有 0.266 的機會出現至少這麼極端的結果。Bonferroni 校正（Bonferroni correction）把每個檢定的顯著水準（significance level）降為 $0.05/17$，不論檢定是否獨立，族系誤差率都不超過 5%。

KNOWLEDGE CHECK 5

A researcher runs 100 independent tests, each of size 5%, and every null is true. How many false rejections are expected, and how likely is at least one?

ANSWER

By (21.6), $100 \times 0.05 = 5$ false rejections are expected, and at least one occurs with probability $1 - 0.95^{100} = 0.994$.

7 HOLM AND THE FALSE DISCOVERY RATE

- [ ] apply Holm's procedure and the Benjamini–Hochberg procedure, and state which error each controls and under which condition

Bonferroni gives every test the same level. Holm's rule starts at the same level and relaxes it after each rejection.

RESULT **Holm's procedure**

Order the p-values, $p_{(1)} \le \cdots \le p_{(m)}$, and let i_{H} be the largest i such that

$$p_{(j)} \le \frac{\alpha}{m - j + 1} \quad \text{for every } j \le i. \tag{21.7}$$

Reject the hypotheses of $p_{(1)}, \ldots, p_{(i_{\mathrm{H}})}$, and none if $p_{(1)}$ misses. Then FWER $\le \alpha$ under any dependence (Holm 1979).

Holm compares $p_{(1)}$ with α/m, then $p_{(2)}$ with $\alpha/(m-1)$, and stops at the first miss. Its levels are never below Bonferroni's.

DEEP DIVE **Why Holm controls the family-wise error rate** optional · click to expand

Suppose Holm rejects a true null, and look at the first true null it rejects, at rank i. The $i - 1$ hypotheses rejected before it are false nulls, so $i - 1 \le m - m_0$, that is $m - i + 1 \ge m_0$. This true null has the smallest p-value among the true nulls, and it passed: that p-value is at most $\alpha/(m - i + 1) \le \alpha/m_0$.

So a false rejection by Holm requires some true null with a p-value at most α/m_0. Each true null's p-value is that small with probability at most α/m_0. By (13.2) the event has probability at most $m_0 \cdot \alpha/m_0 = \alpha$, whatever the dependence.

When many tests run, avoiding every false rejection can cost too many true ones. The **false discovery rate** (FDR) is the expected share of false rejections among all rejections, counted as zero when nothing is rejected.

RESULT **The Benjamini–Hochberg procedure**

Order the p-values and let

$$i_{\mathrm{BH}} := \max\left\{ i : \; p_{(i)} \le \frac{i\,q}{m} \right\}, \tag{21.8}$$

or 0 if no rank passes. Reject the hypotheses of $p_{(1)}, \ldots, p_{(i_{\mathrm{BH}})}$. For independent test statistics the FDR is then at most q (stated without proof; Benjamini and Hochberg 1995).

The rule steps up: it looks for the largest rank that passes, so a p-value above its own level can still be rejected. When every null is true, every rejection is false, and the FDR equals the FWER (Benjamini and Hochberg 1995). SciPy's `false_discovery_control` with `method="by"` (Benjamini and Yekutieli 2001, Theorem 1.3) runs (21.8) with q divided by $\sum_{i=1}^{m} 1/i$ (3.44 for $m = 17$), and controls the FDR under any dependence. LM23 applies these corrections to 28 sector alphas.

EXAMPLE 9

Three rules on ten p-values

Ten tests give the p-values 0.002, 0.0054, 0.0061, 0.011, 0.024, 0.031, 0.09, 0.22, 0.47 and 0.86 (constructed). At $\alpha = q = 5\%$, how many hypotheses do Bonferroni, Holm and Benjamini–Hochberg reject?

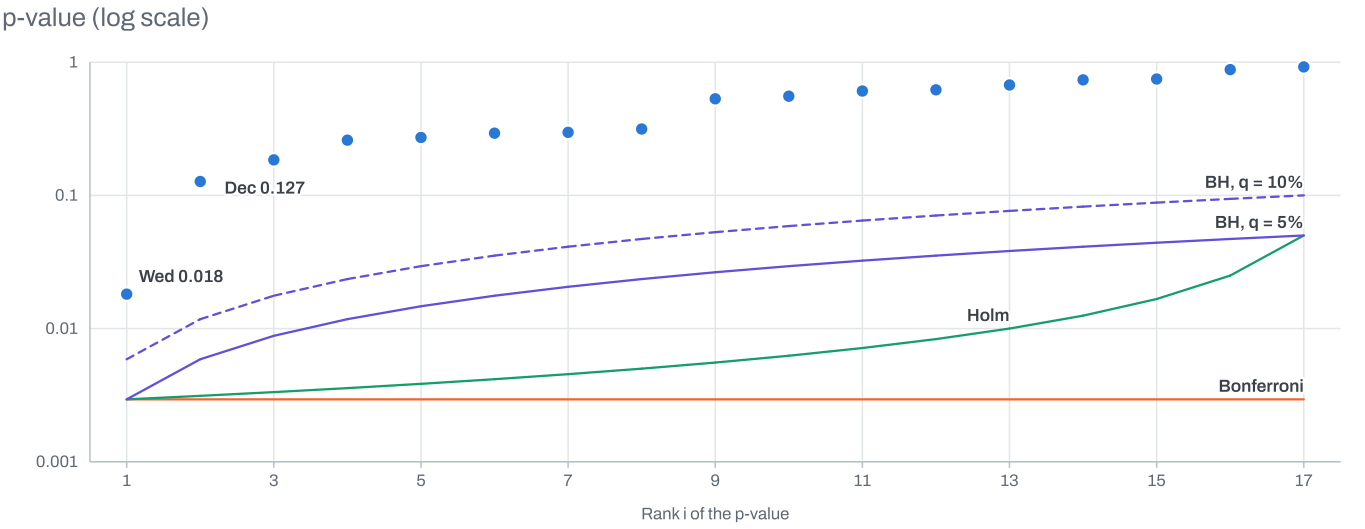
SOLUTION

Bonferroni's level is $0.05/10 = 0.005$, and only 0.002 passes: 1 rejection. Holm's first four levels are 0.005, 0.00556, 0.00625 and 0.00714. The p-values 0.002, 0.0054 and 0.0061 pass, and 0.011 misses: 3 rejections.

Benjamini–Hochberg's levels iq/m run 0.005, 0.010, ..., 0.050. The fifth rank passes, $0.024 \le 0.025$, and every later one misses, from $0.031 > 0.030$ and $0.09 > 0.035$ onwards: 5 rejections.

On the calendar, $p_{(1)} = 0.018$ exceeds 0.00294, the first level of Bonferroni and Holm at 5%, so neither rejects anything. Benjamini–Hochberg steps up, so every rank must be checked: each $p_{(i)}$ exceeds its level iq/m, at least threefold even at $q = 10\%$ (Exhibit 6). Its adjusted p-values, $\min_{j \ge i} m p_{(j)}/j$ (at most 1), start at 0.308.

Exhibit 6: The calendar's p-values against three rules



Note: the 17 p-values of Exhibit 5 in increasing order, on a log scale. Lines: each rule's levels at $\alpha = q = 5\%$, Bonferroni's $\alpha/m = 0.00294$, Holm's $\alpha/(m - i + 1)$ (the second is 0.003125) and Benjamini–Hochberg's iq/m, also at $q = 10\%$. A rule rejects only points on or below its line: Holm from the left until the first miss, Benjamini–Hochberg up to the largest rank on or below. No point qualifies. In-sample. Source: as Exhibit 1.

PITFALL **Choosing the rule after seeing the p-values**

Benjamini–Hochberg rejects at least as many hypotheses as Holm, and Holm at least as many as Bonferroni. Picking the rule that delivers the hoped-for result is one more search: fix the family, the error and the level before the data. Benjamini–Hochberg's guarantee also needs independent statistics. The calendar's 17 are dependent, though the shuffle shows that the FWER barely moves (0.580 against 0.582).

導讀 **偽發現率：控制錯誤的比例**

族系誤差率（family-wise error rate）要求一個都不能錯，檢定（test）一多就太嚴，真正的效果也會被漏掉。偽發現率（false discovery rate）改為控制錯誤拒絕（false rejection）在所有拒絕中所占的比例：拒絕 20 個、錯 1 個，比例是 5%。Benjamini–Hochberg 的方法把排序後第 i 小的 p值（p-value）和 iq/m 比較，在各檢定彼此獨立（independent）時，偽發現率不超過 q。Holm 的方法從最小的 p值往上逐一比較，遇到第一個不通過就停；不論檢定是否獨立，都能控制族系誤差率。月份與星期的 17 個檢定，三種方法都一個也不拒絕。

KNOWLEDGE CHECK 6

Three tests give p-values of 0.004, 0.012 and 0.03. At 5%, how many hypotheses do Bonferroni, Holm and Benjamini–Hochberg reject?

ANSWER

Bonferroni's level is 0.0167, so it rejects 2. Holm compares 0.004, 0.012 and 0.03 with 0.0167, 0.025 and 0.05: all pass, 3 rejections. Benjamini–Hochberg's levels are 0.0167, 0.0333 and 0.05, and the third rank passes: 3 rejections.

8 SEARCHING MANY STRATEGIES: THE HURDLE OF 3

- [ ] explain why the largest t-stat of a search grows with the number of strategies tried, and interpret the t > 3 hurdle

A strategy search is a family too, usually a large one whose members go unreported. Model it as m tests whose null statistics are independent and standard normal. The largest of them exceeds a level when any one does, and (16.1) multiplies the probabilities that none does.

RESULT **The largest of m null statistics**

For independent standard normal $Z_1, \dots, Z_m$ and every x,

$$\Pr\left(\max_i Z_i \le x\right) = \Phi(x)^m. \tag{21.9}$$

The largest has median $\Phi^{-1}(0.5^{1/m})$ and u quantile $\Phi^{-1}(u^{1/m})$.

The median of the best grows with m: 1.75 at $m = 17$, 2.46 at 100, 3.20 at 1,000 and 3.81 at 10,000. At $m =$ 1,000 its middle 90% runs from 2.75 to 3.88, and it exceeds 3 with probability 0.74 (Exhibit 7).

EXAMPLE 10

Ten thousand random rules on TAIEX

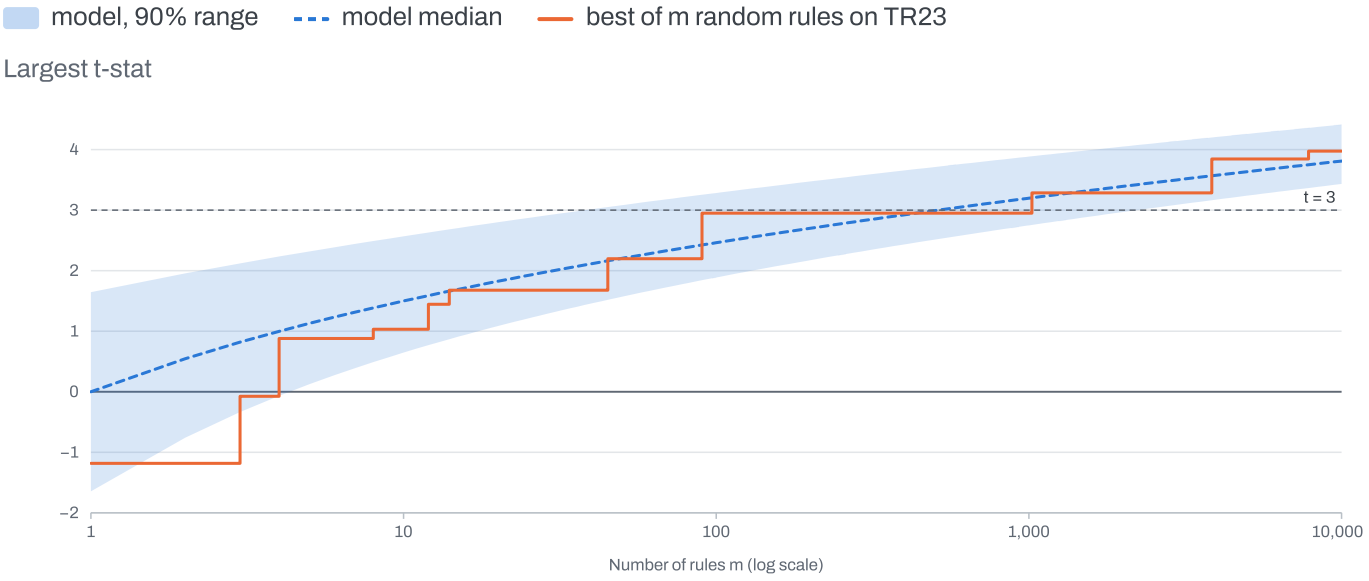
Experiment 10 draws every position in one call on `default_rng([10, 20261011])`, `integers(0, 2, size=(10000, T), dtype=np.int8)`, with 1 for long and 0 for short. Rule i of the 10,000 is long or short one unit each day by this fair coin, $f_{i,t} = \pm 1$. Its daily return is $f_{i,t}R_t$, and (21.1) gives its t-stat with $\mu_{\text{null}} = 0$. How large is the best t-stat?

SOLUTION

The 10,000 t-stats average −0.01 (standard error 0.01), with a standard deviation of 0.995 (batch standard error 0.007, from 20 batches of 500, LM18). They behave like standard normal draws. 13 reach 3, against 10,000 $\Phi(-3) = 13.5$ expected.

The best after 100 rules is 2.95 (rule 90), and it is still 2.95 after 1,000. Rule 1,022 reaches 3.28, and the best of all 10,000 is 3.97 (rule 7,840). This is one search, a single draw with no standard error. Under (21.9) the best of 1,000 has median 3.20 and exceeds 3 with probability 0.74. Model on TR23's returns.

Exhibit 7: The best t-stat of a search



Note: number of rules m on a log axis. Model: the median and the 5% to 95% range of the largest of m independent standard normal statistics, $\Phi^{-1}(u^{1/m})$ by (21.9). Step line: the best t-stat (21.1) among the first m random rules of Example 10, through its 11 records; one search, not an average. Source: as Exhibit 1.

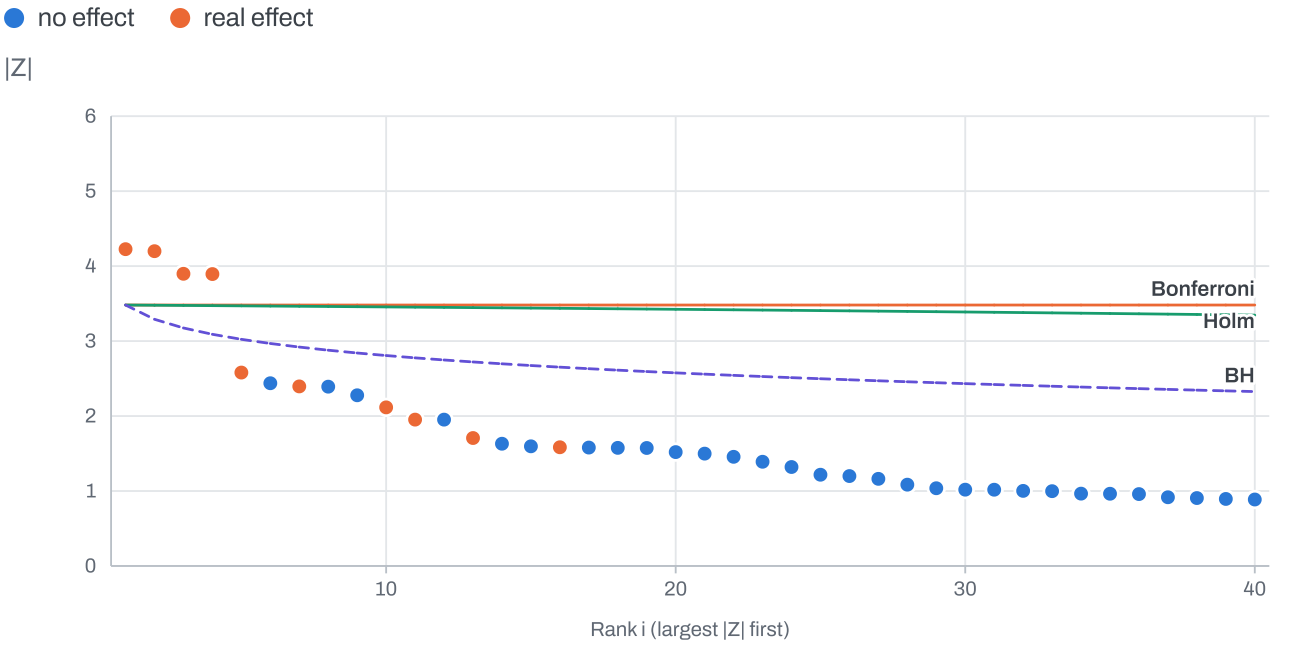
Harvey, Liu and Zhu (2016) argue that a new factor needs "a t-statistic greater than 3.0". The two-sided p-value at $|\text{t-stat}| = 3$ is $2\Phi(-3) = 0.0027$, Bonferroni's level for about 18.5 tests at 5%. So a hurdle of 3 corrects for a search of about 18.5 tests, while the median best of 1,000 null rules already passes it. TR23's mean clears both bars: its t-stat, 3.96, exceeds 3 and Bonferroni's 3.134 for this module's 29 tests.

LM4 warned that backtests overstate Sharpe ratios. LM33 deflates a Sharpe ratio for the number of trials, and LM34 tests the best rule of a search on later data (White 2000; Romano and Wolf 2005).

The widget applies the three rules to a family of m normal statistics, some with a real effect. It starts with 100 tests at the 5% level, 10% of them with a mean t-stat of 3.

Many tests, three rules

INTERACTIVE



Bonferroni	Holm	Benjamini–Hochberg
4 rejected, 0 false	**4 rejected, 0 false**	**4 rejected, 0 false**
No correction, $p \leq \alpha$	Largest |Z| without an effect	Draw (seed)
10 rejected, 3 false (4.50 expected)	**2.44**	**1**

Each draw gives m standard normal statistics from the book's seeded generator, and adds the chosen mean t-stat to the first share of them: those have a real effect, the rest none. The chart shows the 40 largest |Z| by rank, with each rule's level; a rule rejects the leading ranks (Holm until its first miss, Benjamini–Hochberg up to its largest rank on or below its line). New draw moves to the next seed. The widget's generator is not NumPy's; the text quotes no widget value.

導讀 **資料偵伺：最好的那一個為什麼不可信**

資料偵伺（data snooping）是在同一份資料上試很多策略，只報告最好的那一個。就算每個策略都毫無預測力，最好的 t 統計量（t-statistic）也會隨嘗試的數目變大：在獨立（independent）的情形下，1,000 個裡最好的中位數（median）是 3.20。在加權股價報酬指數上用擲硬幣決定每天作多（long）或放空（short），10,000 條隨機規則中最好的 t 統計量達到 3.97。Harvey、Liu 和 Zhu 主張新因子（factor）的 t 統計量要超過 3，這只相當於對 18.5 個檢定（test）做 Bonferroni 校正（Bonferroni correction）。看到漂亮的回測（backtest），先問它試了幾個。

KNOWLEDGE CHECK 7

How many independent null strategies give an even chance that the best t-stat exceeds 3?

ANSWER

By (21.9) we need $1-\Phi(3)^m \geq 0.5$, that is $m \geq \ln 0.5 / \ln \Phi(3) = 513.1$: 514 strategies.

9 TESTS IN PYTHON

- [ ] run these tests in Python by hand and with SciPy and statsmodels, checking each function's defaults

NumPy computes each statistic, and SciPy supplies the tails.

PYTHON

```python
import numpy as np
import polars as pl
from scipy import stats
from statsmodels.stats.multitest import multipletests

tr = pl.read_csv("data/taiex_tr.csv")["price"].to_numpy()
R = tr[1:] / tr[:-1] - 1                    # T = 5,849 simple returns
T = len(R)
tstat = R.mean() / (R.std(ddof=1) / np.sqrt(T))   # (21.1): 3.96
print(2 * stats.norm.sf(abs(tstat)))         # 7.6e-05, normal
print(2 * stats.t.sf(abs(tstat), T - 1))     # 7.7e-05, t with T - 1
print(stats.binom.sf(3205 - 1, T, 0.5))      # 1.2e-13, up days
print(stats.chi2.sf(2.69, 1))                # 0.10, signs, (21.5)
p = np.array([0.002, 0.0054, 0.0061, 0.011, 0.024,
              0.031, 0.09, 0.22, 0.47, 0.86])
for method in ("bonferroni", "holm", "fdr_bh"):  # 1, 3, 5
    print(method, multipletests(p, alpha=0.05, method=method)[0].sum())
print((stats.false_discovery_control(p) <= 0.05).sum())  # 5, BH
```

Check each function's defaults. `multipletests` defaults to Holm–Šidák (`method="hs"`), not Holm, so always pass `method`. Holm–Šidák, for independent tests, runs Holm's steps at Šidák's levels $1-(1-\alpha)^{1/(m-i+1)}$ in place of $\alpha/(m-i+1)$. SciPy's `ttest_ind` pools the two variances unless given `equal_var=False`. Its `chi2_contingency` applies Yates' correction to a 2×2 table unless given `correction=False`. `jarque_bera` uses moments with divisor T. NumPy's `std` divides by T and Polars' by $T-1$ unless given `ddof`.

IN PYTHON **Companion lab**

The Polars-first notebook `LM21_lab.ipynb` rebuilds the module's exhibits and computational examples from TR23's closes, runs experiments 8 and 10, and solves the computational practice problems. It ends with the calendar family in each limit regime under Holm and Benjamini–Hochberg.

The notebook reads FinMind files that you download once with your own token. From the repository root, run `uv run python data/fetch_finmind.py`, then `uv run python data/make_lab_data.py`. FinMind's licence does not allow the book to redistribute the data. After that, the notebook runs offline.

KNOWLEDGE CHECK 8

A t-stat of 2 comes from $T = 30$ returns. Compare the two-sided p-values from t_{29} and from the normal. Why do Example 3's two p-values differ by only about 1%?

ANSWER

`2 * stats.t.sf(2, 29)` gives 0.055 and `2 * stats.norm.sf(2)` gives 0.0455: at $T = 30$ the heavier tails of t_{29} matter. At $T = 5{,}849$, t_{5848} is nearly normal, so the p-values are 7.6×10^{-5} and 7.7×10^{-5}.

SUMMARY

- A test fixes H_0, H_1 and a level before the data; the p-value (21.2) is a probability of data given H_0.
- TR23's mean has a t-stat of 3.96, (21.1); the regimes' difference has only 1.08, (21.3).
- Power grows with $\mathrm{SR}\sqrt{\tau}$, (21.4): an annual Sharpe ratio of 0.5 needs 31.4 years for 80% power.
- The 2×2 χ^2 is $n\hat{\rho}^2$, (21.5); clustered days break the null laws of count tests.
- A family expects false rejections, (21.6). Bonferroni and Holm (21.7) control the FWER under any dependence, Benjamini–Hochberg (21.8) the FDR for independent statistics.
- The largest of m null t-stats grows with m, (21.9), and a hurdle of 3 corrects for about 18.5 tests.
- Report every test with its family: this module runs 29 on TR23.

PRACTICE PROBLEMS

1. A test at the 5% level rejects H_0. Which statement is true?
   - **A.** The probability that H_0 is true is below 5%.
   - **B.** If H_0 were true, a statistic at least this extreme would occur with probability at most 5%.

C. The probability that H_1 is true is at least 95%.

2. Twenty independent tests each have size 5%, and every null is true. The probability of at least one rejection is *closest* to:

   A. 0.05

   B. 1.00

   C. 0.64

3. Returns are iid with known σ. For 80% power against an annual Sharpe ratio of 0.5, two-sided at 5%, the years of daily data needed are *closest* to:

   A. 31.4

   B. 15.4

   C. 5.60

4. In 4,000 day pairs two indicators correlate $\hat{\rho} = 0.03$. The 2×2 χ^2 of (21.5) is *closest* to:

   A. 120

   B. 3.60

   C. 1.90

5. Five tests give p-values of 0.003, 0.012, 0.020, 0.030 and 0.40. At $q = 5\%$, how many hypotheses does Benjamini–Hochberg reject?

   A. 4

   B. 2

   C. 1

6. A hypothetical manager beat the index in 9 of 12 months. Test $\theta = \frac{1}{2}$ against $\theta > \frac{1}{2}$ at 10%. (a) Find the p-value. (b) Find the rejection region and its size. (c) Find the power at $\theta = 0.55$. (d) At 5%, with $\Phi^{-1}(0.95) = 1.644854$ and $\Phi^{-1}(0.8) = 0.841621$, find by the normal approximation, using $\hat{\theta}$'s standard deviation under each hypothesis, the months that give 80% power at $\theta = 0.55$. Check them with the exact binomial.

7. LM15's 292 blocks of 20 sessions (TR23's first 5,840 returns) have up-day counts with variance 4.72 (divisor 292). The binomial gives 4.95, and a model with next-day correlation 5.16. Use the standard error of a variance of n nearly normal draws, $\sigma_X^2\sqrt{2/n}$, with σ_X^2 the model's value. It is the square root of LM20's $\mathrm{Var}(s^2) \approx (\mathrm{Kurt} - 1)\sigma^4/T$ at $\mathrm{Kurt} = 3$, with n draws in place of T. (a) Test 4.72 against each model value, two-sided at 5%. (b) How many blocks would detect the gap of 0.21 at $z = 1.96$, with the binomial's standard error? Give the sessions and the years at $A = 246.12$.

8. On LM16's 4,382 U28 days, Semiconductor fell 1% or more on 859, FinancialInsurance on 538, and both on 347. (a) Build the 2×2 table, and compute $\hat{\rho}$ and the χ^2 of (21.5). (b) Is the χ^2_1 law credible here, and does the conclusion survive?

9. SciPy's `jarque_bera` computes

$$\mathrm{JB} = \frac{T}{6}\left(\mathrm{Skew}^2 + \tfrac{1}{4}(\mathrm{Kurt}-3)^2\right)$$

with T moments of divisor , χ^2_2 after Jarque and Bera (1980). For iid normal returns JB is approximately (stated LM14 without proof). (a) From 's skewness −0.34 and excess kurtosis 5.27 for TR23, compute JB and its p-value by (16.9). (b) What does the rejection say?

10. *(Python)* Compute the calendar family's statistics (21.3) by hand and their two-sided normal p-values. Apply `multipletests` with `method="holm"`, with `method="fdr_bh"` and without a method, and `stats.false_discovery_control`. Give the rejections at 5% and Benjamini–Hochberg's two smallest adjusted p-values.

11. *(Python)* Draw 10,000 families of 17 independent standard normal statistics, `default_rng([11, 20261011]).standard_normal((10000, 17))`. Find the share of families with a rejection at $|z| \geq 1.96$ and at Bonferroni's 2.974, with standard errors by (18.2), against the exact values.

SOLUTIONS

1. **B** is correct: it restates the p-value, a probability of data given H_0. A reads it as a probability of H_0, which needs a prior (LM24). C makes the same error with H_1.

2. **C** is correct: $1 - 0.95^{20} = 0.64$ by (13.7). A is one test's size. B is the expected number of false rejections, $20 \times 0.05 = 1.00$, which (13.2) makes only a bound on the probability.

3. **A** is correct: $\tau = ((1.959964 + 0.841621)/0.5)^2 = 31.4$ by (21.4). B, $(1.959964/0.5)^2 = 15.4$, drops $\Phi^{-1}(0.8)$, which gives power of about 50%. C, $2.801585/0.5 = 5.60$, forgets to square.

4. **B** is correct: $4{,}000 \times 0.03^2 = 3.60$, a p-value of 0.058, so independence is not rejected at 5%. A is $n\hat{\rho} = 120$. C is $\sqrt{n}\,\hat{\rho} = 1.90$, the z of (19.6), not its square.

5. **A** is correct. The levels iq/m are 0.01, 0.02, 0.03, 0.04 and 0.05; the fourth rank passes ($0.030 \leq 0.04$) and the fifth misses, so 4. B is Holm's count: 0.003 and 0.012 pass 0.01 and 0.0125, and 0.020 misses 0.0167. C is Bonferroni's: only 0.003 passes 0.01.

6. (a) The p-value is $\Pr(X \geq 9) = 299/4{,}096 = 0.0730$, significant at 10%. (b) The tail $\Pr(X \geq 8) = 0.194$ exceeds 0.10, so the region is $X \geq 9$, with size 0.0730: the record lies on the region's edge, so its p-value is the size. (c) Under $\mathrm{Bin}(12, 0.55)$, $\Pr(X \geq 9) = 0.134$. (d) The test rejects when $\hat{\theta} \geq \frac{1}{2} + 1.644854 \times 0.5/\sqrt{n}$. Under $\theta = 0.55$, $\hat{\theta}$ has standard deviation $\sqrt{0.55 \times 0.45/n}$, so 80% power needs

$$\begin{aligned} \sqrt{n} \times 0.05 &= 1.644854 \times 0.5 \\ +0.841621\sqrt{0.55 \times 0.45} &= 1.241128, \\ n = (1.241128/0.05)^2 &= 616.16. \end{aligned}$$

   So 617 months, or 51.4 years. Exactly, at $n = 617$ the 5% test rejects at $X \geq 330$, with size 0.0454 and power 0.7874, just short of 80%. Exact power is not monotone in n: it first reaches 80% at 620 months (reject at $X \geq 331$, size 0.0498, power 0.8018).

7. (a) From the printed values,

$$z = \frac{4.72 - 4.95}{4.95\sqrt{2/292}} = -0.56,$$
$$z = \frac{4.72 - 5.16}{5.16\sqrt{2/292}} = -1.03;$$

unrounded inputs give −0.58 and −1.03. Neither is significant at 5%. (b) The gap 0.21 equals $1.959964 \times 4.95\sqrt{2/n}$ at

$$n = 2\Big(\frac{1.959964 \times 4.95}{0.21}\Big)^2 = 4{,}268.7,$$

so 4,269 blocks, 85,380 sessions or 347 years: 292 blocks cannot see a dependence this small.

8. (a) The cells are 347, $859 - 347 = 512$, $538 - 347 = 191$ and $4{,}382 - 347 - 512 - 191 = 3{,}332$. Then $\hat\rho = 0.423$ and $\chi^2 = 4{,}382 \times \hat\rho^2 = 784$. The ratio $347 \times 4{,}382/(859 \times 538)$ recomputes LM16's 3.29 as 3.2902. (b) Sector returns cluster, so the pairs are not iid and χ^2 spreads more than χ^2_1. Yet clustering would have to widen the spread of $\sqrt{n}\,\hat\rho$ 14.3-fold, its variance 204-fold, to bring 784 down to the 5% point 3.84. In-sample, from LM16's printed U28 counts.

9. (a) The statistic is

$$\mathrm{JB} = \frac{5849}{6}\Big(0.34^2 + \tfrac{1}{4} \times 5.27^2\Big) = 6{,}881.$$

By (16.9), the χ^2_2 tail beyond JB is $e^{-\mathrm{JB}/2}$, about 10^{-1494}; unrounded moments give 6,875 and 10^{-1493}. (b) The test rejects iid normal returns beyond doubt: TR23's fat tails (LM14, LM15) are no sampling accident. It does not say which model fits, and at this T even a small departure rejects.

10. Holm and Holm–Šidák stop at the first rank: $p_{(1)} = 0.018$ exceeds their first levels, 0.00294 and $1 - 0.95^{1/17} = 0.00301$. Benjamini–Hochberg, in `multipletests` and in `false_discovery_control`, rejects nothing either, because every $p_{(i)}$ exceeds iq/m. Without a method, `multipletests` runs Holm–Šidák. The two smallest Benjamini–Hochberg adjusted p-values are 0.308 and 0.669.

11. The share with a rejection at $|z| \geq 1.96$ is 0.590 ± 0.005, against the exact $1 - 0.95^{17} = 0.582$. With Bonferroni it is 0.051 ± 0.002, against the exact $1 - (1 - 0.05/17)^{17} = 0.04884$. Both lie within two standard errors of the exact values. Model.

GLOSSARY

Term	中文	Meaning
Hypothesis test	假設檢定	A rule that rejects or keeps a null hypothesis from data, at a fixed level.
Null hypothesis	虛無假設	H_0, the parameter value under test.
Alternative hypothesis	對立假設	H_1, where the parameter lies if H_0 is false; one- or two-sided.
Significance level	顯著水準	α, the largest allowed probability of a type I error.
p-value	p值	The probability under H_0 of a statistic at least as extreme as the one observed, (21.2).
Type I error	第一型錯誤	Rejecting a true null hypothesis.
Type II error	第二型錯誤	Keeping a false null hypothesis.
Power	檢定力	The probability of rejecting H_0 under a given alternative, (21.4).
Multiple testing	多重假設檢定	Running many tests on the same data.
Family-wise error rate	族系誤差率	The probability of at least one false rejection in a family, (21.6).
Bonferroni correction	Bonferroni 校正	Running each of m tests at level α/m, (21.6).
False discovery rate	偽發現率	The expected share of false rejections among all rejections, zero if none, (21.8).
Data snooping	資料偵伺	Searching many models or rules on one data set and reporting the best, (21.9).

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